# Unseen Strings, Familiar Tasks: Certified Semantic Exposure in Procedural Reasoning Benchmarks

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## Abstract

Many reasoning benchmarks use a program to generate their tasks, and they build the test set by holding out a representation: a parameter setting, a rule string, or a program that never appeared in training. We show that such splits can fail silently, because different representations can produce the same observable behavior. Using tools that produce a proof for every match, we audit three public protocols and find three distinct failures. In SLR-Bench, none of the 1,000 test rules repeats a training string, yet we certify that at least 45.2% are logically equivalent to a training rule, with a checkable proof for every pair; the true overlap may be higher. In Tensor Logic, the split removes test edges from training but breaks many of the paths that were meant to justify the test queries, leaving between 484 and 1,000 of the 1,000 test queries with a matching path still present in training. In the Occam in-context-learning benchmark, the generator keeps sampling from a small hidden pool of tasks: at least two of the twenty evaluation tasks reappear in training, and the true number may be as high as four. We then asked whether seeing a task in training actually changes what a model scores. The answer depends on the setting. Across three Boolean in-context-learning tasks, removing exact exposure changes mean accuracy by at most 0.09 points, even where exposure reaches 93.44%; a controlled six-architecture study finds a small gain (+1.8 points); and a preregistered experiment on the public CLUTRR kinship benchmark, confirmed on fresh targets, finds a gain of 23.3 points (95% CI $[17.7, 29.2]$). Our tools are simple in spirit: say what "the same task" means before measuring, grade how strong each piece of evidence is, and report an interval instead of treating missing evidence as no overlap. Syntactic novelty is not semantic novelty, and exposure is not exploitation; benchmark protocols should measure both.

**Keywords:** benchmark auditing, semantic novelty, data contamination, evaluation methodology, preregistration

## 1 Introduction

SLR-Bench is a published logical-reasoning benchmark whose test split is built to be new: none of its 1,000 test rules repeats any of the 18,053 training rule strings (Helff et al., 2026). Under the benchmark's own protocol, every test rule counts as an unseen task. We show that this guarantee does not hold. Using a tool that rewrites rules into a standard form and proves every match, we certify that at least 452 of the 1,000 test rules—45.2%—are logically equivalent to a training rule, and the true overlap may be higher (Section 6, Figure 1). The overlap is invisible to the split's own criterion because the equivalent rules differ as strings: variables are renamed, conditions reordered, redundant requirements dropped.




Figure 1: SLR-Bench `v1-All`. The split holds out rule strings, and string matching finds zero overlap (left bar). Token-level and proof-producing semantic comparisons tell a different story, certifying a 45.2% lower bound on train–test overlap. Aggregate, negated, arithmetic, and unresolved disjunctive rules are not counted, so the final bar is a floor, not an estimate of total overlap.

Generated benchmarks are popular testbeds—for in-context function learning (Garg et al., 2022; Akyürek et al., 2023; Bhattamishra et al., 2024), program induction (Reed and de Freitas, 2016; Zaremba and Sutskever, 2014), and systematic generalization (Lake and Baroni, 2018)—because the rules behind the data are known and new data can be sampled without manual annotation. But most of their splits hold out a *representation*: a parameter setting, a rule string, a program. The evaluation claim, meanwhile, is about a *task*: the observable behavior that the representation implements. Because different representations can implement the same behavior, a representation-level split cannot by itself certify a task-level novelty claim. We find that this gap opens in three distinct ways across three public protocols: *semantic aliases*, where different rules induce the same logic (SLR-Bench); *witness breakage*, where the mutation that hides test edges also deletes edges used by the split's own witnesses (Tensor Logic: 379 of 1,000 designated test paths survive, and between 484 and 1,000 of the 1,000 test queries still have a matching path in training (Shah and Zadrozny, 2026)); and *finite-space resampling*, where an online generator keeps drawing from a small hidden task pool (in the Occam benchmark, at least two of twenty evaluated tasks reappear in training, the true number may be four, and at least 93.7% of the 800,000 training episodes reuse an earlier hidden task (Deora et al., 2025)).

Does such exposure actually change what models score? The tempting answer is yes, but it must be measured, and it is not uniform. We ran three controlled studies in which paired training runs differ only in whether the target tasks are already seen during training. Across three Boolean in-context-learning tasks, removing exact exposure changes mean accuracy by at most 0.09 percentage points, even where 93.44% of the evaluation functions appear in training. A controlled six-architecture study finds a small gain (+1.8 points). A preregistered experiment on the public CLUTRR kinship benchmark, confirmed on fresh targets, finds a gain of 23.3 points (95% CI $[17.7, 29.2]$) (Sinha et al., 2019). Exposure

and exploitation are different things, and a credible benchmark audit must report them separately.

Making such audits possible takes three simple tools. An *observation contract* says, before any measurement, what "the same task" means for a given claim. An *evidence grade* marks every overlap finding as exact, a sound lower bound, estimated, or unresolved—and unresolved items are never counted as novel. A *certified exposure interval* turns an incomplete search for equivalent items into explicit bounds $[L, U]$ instead of silently counting them as zero. An open toolkit checks every reported claim against its declared unit, evidence grade, and source, and rebuilds the analyses in this paper from saved result files.

**Contributions.**

- **Findings.** Certified audits of three public protocols: a sound 45.2% lower bound on semantic overlap for SLR-Bench despite zero string-level overlap; witness breakage and endpoint exposure in $[484, 1000]/1000$ for Tensor Logic; and certified exposure of $[2, 4]/20$ tasks under finite-space resampling in the Occam benchmark.
- **Behavioral evidence.** Three controlled studies that separate exposure from exploitation, spanning a near-zero result (Boolean), a small gain (six architectures), and a large gain (CLUTRR, preregistered, confirmed on fresh targets).
- **Instruments.** Observation contracts, evidence grades, and certified exposure intervals (Propositions 1 and 2), with an open, schema-checked toolkit.
- **Guidance.** A short reporting standard for benchmark authors: name the held-out object, grade the evidence, report the interval, and measure behavior separately (Section 7).

This is an evaluation-methodology study, not a new model or a general procedure for program equivalence. Its strongest conclusion is deliberately bounded: training curricula that differ only in semantic exposure can produce large differences in final accuracy, but exposure alone does not reveal how a model obtained its prediction.

## 2 What Counts as a New Task?

### 2.1 Representation, Task, and Family

A generator parameter setting, a rule string, or a program is a *representation* $r$, drawn from a space $\mathcal{R}$. An evaluation claim, however, is rarely about the representation itself; it is about the behavior the representation implements. To make "behavior" precise, we fix an *observation contract* $\mathcal{O}$ before any measurement: a written agreement on which inputs are allowed, which outputs are observed, how termination, errors, and nondeterminism are treated, and nothing else. Under a contract, each representation induces a semantic object through

$$q_{\mathcal{O}} : \mathcal{R} \to \mathcal{F}_{\mathcal{O}}. \tag{1}$$

For the finite, deterministic, effect-free tasks studied here, $\mathcal{F}_{\mathcal{O}}$ is simply the space of complete input–output functions, and two representations are the *same task* iff $q_{\mathcal{O}}(r) = q_{\mathcal{O}}(r')$. We do

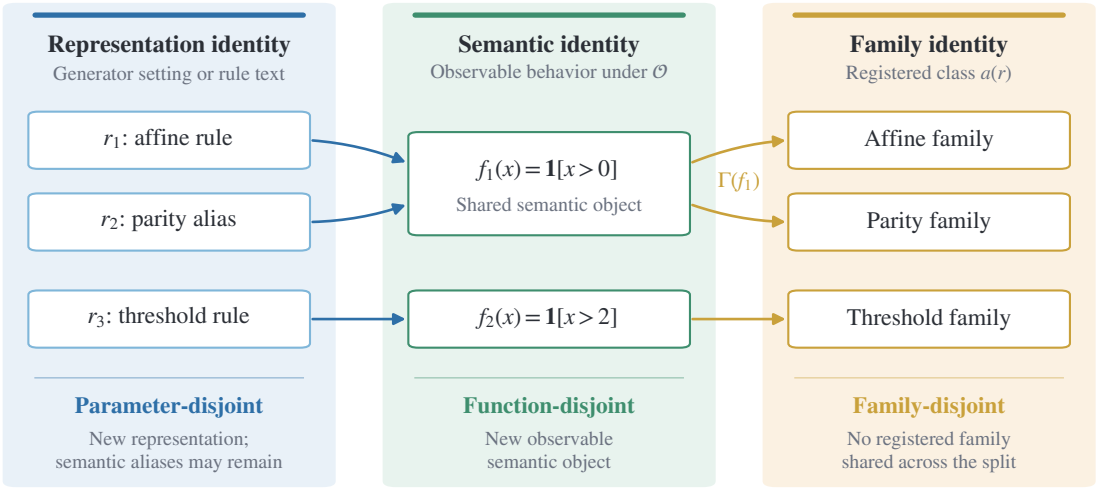



Figure 2: Three levels of novelty a split can claim. A representation has a registered family and, under an observation contract, induces a task—an observable behavior. Several representations, including ones from different families, can induce the same task. Holding out parameters, holding out behaviors, and holding out families are therefore different promises.

not assume that general program equivalence is decidable; the contract only has to support the claim at hand.

A representation also carries a family label $a(r) \in \mathcal{G}$, such as "affine rule" or "parity rule." Families classify representations, not tasks: one task can be implemented by representations from several families. We write

$$\Gamma(f) = \{a(r) : q_{\mathcal{O}}(r) = f\}. \tag{2}$$

for the set of families that realize task $f$; the case $|\Gamma(f)| > 1$ is exactly the cross-family aliasing that our first behavioral study exploits (Section 4.1).

Figure 2 shows three things a split can hold out, and they are not interchangeable. **Parameter-disjoint** evaluation holds out $r$ itself but may keep an alias $r'$ with $q_{\mathcal{O}}(r') = q_{\mathcal{O}}(r)$. **Function-disjoint** evaluation holds out the behavior:

$$q_{\mathcal{O}}(\mathcal{R}_{\text{train}}) \cap q_{\mathcal{O}}(\mathcal{R}_{\text{test}}) = \varnothing. \tag{3}$$

**Family-disjoint** evaluation holds out the family label:

$$a(\mathcal{R}_{\text{train}}) \cap a(\mathcal{R}_{\text{test}}) = \varnothing. \tag{4}$$

Neither of the last two implies the other: two different tasks can share a family, and two different families can implement the same task.

The *semantic unit*—what counts as one task—must fit the claim and be fixed before overlap is measured. We ask for a prospective test: the claim names what should stay

fixed and what should vary, then supplies *must-match* pairs (items that should count as the same task) and *must-differ* pairs (items that should not). A unit fails if it splits a must-match pair or merges a must-differ pair; when several defensible units pass and disagree, we report all of them rather than assert invariance. Units differ by benchmark: a finite rule can be identified by its complete output vector, a logic program by the relation it defines, CLUTRR by the ordered relation composition plus the target relation rather than the story string, and graph reasoning by endpoint-conditioned paths or relation schemas. Table 1 shows why this matters: Tensor Logic has endpoint exposure $[484, 1000]/1000$ but schema exposure $[999, 1000]/1000$, while SLR string identity gives zero overlap and the registered unit gives a 45.2% sound lower bound; a single pooled "contamination rate" would erase these distinctions.

### 2.2 Evidence Grades

Not every overlap finding is equally strong, and we keep the strength attached to the finding. A complete truth table or a provably injective identifier gives an *exact* test; a normalization that emits a proof for every match gives a *sound lower bound* (every collision it reports is real, but items it cannot resolve may hide more); approximate execution signatures give an *estimate* and need an error analysis. An unresolved item is never evidence of novelty. The grades describe identification conditional on our machinery, not bugs inside it, so assurance is recorded separately per audit: contract, source revisions and hashes, semantic-ID implementation, soundness argument, validation coverage, known failure modes, and qualitative residual risk (Table 16; Appendix K).

### 2.3 Training-Budget Exposure

Splits are not the only place novelty leaks: an online generator that draws fresh episodes can still reuse tasks from a finite pool. For a discrete function distribution $p(f)$ and $N$ independent training draws, the probability that an independent test function was seen during training is

$$P_{\text{seen}}(N) = \sum_f p(f) \left[1 - (1 - p(f))^N\right]. \tag{5}$$

The common shortcut $N/|\mathcal{F}|$ is valid only when sampling is uniform and coverage is low. A nonuniform sampler running a long stream makes common functions almost certain to repeat while rare ones stay novel; the Occam audit in Section 6 is exactly this failure.

## 3 Auditing With Incomplete Evidence

### 3.1 The Audit Workflow

Our audits follow four steps. **First, declare the held-out object**: text, parameters, complete functions, or families. **Second, build and grade a semantic signature**: exact maps when the space is finite, proof-producing normalization when possible, explicitly estimated signatures otherwise. **Third, measure overlap under the actual sampling budget**, including online training streams, not only the nominal split. **Fourth, separate**

**validity from impact**: check whether the protocol supports its novelty claim before asking whether the exposure changed model behavior.

### 3.2 When the Witness Search Is Incomplete, Report an Interval

Equivalence search is often incomplete: some items carry a certificate, others do not. The temptation is to count the uncertified ones as novel. That is wrong, and the fix is to report exactly what the certificates imply. Let $z_i$ indicate whether evaluation task $i$ occurred in training. A verified certificate on a group $G_j$ supplies $\ell_j \le \sum_{i \in G_j} z_i \le u_j$. For pairwise-disjoint certificate groups and uncovered set $R$, Proposition 1 gives

$$\boxed{\sum_j \ell_j \le C(E) \le |R| + \sum_j u_j}. \tag{6}$$

**Proposition 1 (Certified exposure interval)** *Under a frozen finite denominator, deterministic semantic identities, sound group certificates, pairwise-disjoint groups, and no constraints on uncovered items beyond binary membership, Equation 6 contains every exposure count consistent with the certificates, and both endpoints are attained.*

The proof is elementary, and we mean that as a feature: this is an accounting device, not new mathematics. The lower endpoint puts exactly $\ell_j$ exposed items in each group and none in $R$; the upper puts $u_j$ in each group and all of $R$. No tighter interval follows from those certificates alone (proof, a grouped-seed corollary, and falsification tests are in Appendix J). Operationally, an audit must report $[L, U]$, and rows without witnesses stay out of the "novel" column.

When certificate groups overlap the closed form no longer applies; the toolkit either solves the binary program exactly by mixed-integer linear programming or returns a linear-programming outer interval with independently checkable dual certificates (Appendix J.4).

A downstream question asks how well the truly novel subset could be doing among unresolved items with observed scores; Proposition 2 gives the sharp answer.

**Proposition 2 (Novelty-conditioned score interval)** *Let $\mathcal{R}$ be $M$ unresolved items with observed scores $y_i \in [0,1]$, sorted as $y_{(1)} \le \cdots \le y_{(M)}$, and suppose at most $U < M$ of them are exposed. Then the mean score on the truly novel items lies in*

$$\left[ \frac{1}{M-U} \sum_{k=1}^{M-U} y_{(k)},\ \frac{1}{M-U} \sum_{k=U+1}^{M} y_{(k)} \right],$$

*both endpoints are attained by some admissible novel set, and the interval widens monotonically in $U$.*

This is a trimming bound in the sense of Lee (2009) and Horowitz and Manski (1995): without a nontrivial upper bound on exposure, any nonempty subset could be the novel one and the interval is $[\min_i y_i, \max_i y_i]$ (sensitivity to $U$ and proof in Appendix J.3). The bound isolates witness incompleteness only: it assumes correct sources, semantic IDs, and certificate implementations, and does not absorb machinery failures, so it is read together with the assurance record.

For compactness, the summary table abbreviates confidence interval (CI) and in-context learning (ICL).

Table 1: Cross-benchmark evidence summary. Semantic units differ by row and must not be pooled into a prevalence estimate.

| Case | Nominal holdout | Measured semantic exposure | Behavioral evidence |
|---|---|---|---|
| Exact-alias intervention | Cross-family representation | Complementary exposure of 40 exact functions | +1.832 pp macro, 95% CI $[1.470, 2.189]$ |
| Boolean ICL | Fresh function draw | 0.81–93.44% exact exposure | −0.06 to +0.09 pp; below preregistered threshold |
| CLUTRR | Disjoint source stories and permutations | Randomized exposure of relation-composition patterns | V4: +23.339 pp, 95% CI $[17.750, 29.226]$ |
| SLR-Bench | Raw rule string | $\geq 45.2\%$ sound logical overlap | Identification stopped before outcome fitting |
| Tensor Logic | Direct-edge removal | $[484, 1000]/1000$ endpoint; $[999, 1000]/1000$ schema | No reproducible checkpoint or per-item outputs |
| Occam Markov | Unseen transition task | $[2, 4]/20$ exact tasks per main experiment | Aggregate plots only; regression scope unavailable |

## 4 Does Exposure Change Behavior? Two Controlled Studies

Knowing that a test task was exposed does not say whether the exposure helped. We therefore ran controlled studies with frozen endpoints and paired training runs that differ only in which target tasks are exposed. This section reports two of them, with opposite outcomes: a small positive gain in a controlled six-architecture study, and a near-zero result in a Boolean in-context-learning task despite almost complete exposure. Section 5 reports the third, a large gain on the public CLUTRR benchmark. The studies share a design idea but deliberately no pooled effect estimate.

### 4.1 Study 1: Exact Aliases in a Controlled Generator

Our finite rule generator has 560 parameter identities but only 518 complete functions: exactly 40 functions are implemented twice, once by an affine rule and once by a parity rule. This is the cleanest possible testbed for aliasing. We pair the two identities of each aliased function ($f_{1,b}$ with $f_{11,b}$ for each intercept $b \in \{0, \ldots, 19\}$). For each model seed, a frozen SHA-256 bit assigns one member of every pair to Arm A and the other to Arm B, so every target is exposed in exactly one arm and unexposed in the other. Both arms share 478 common base functions, equal family counts, and 20,000 paired episodes; initialization, episode order, and optimizer schedule are identical.

We train six architectures—Deep Sets (Zaheer et al., 2017), Relation Network (Sung et al., 2018), Set Transformer (Lee et al., 2019), modern Hopfield retrieval (Ramsauer et al., 2021), Perceiver (Jaegle et al., 2021), and Transformer (Vaswani et al., 2017)—with five seeds and two arms, giving 60 fixed epoch-60 runs, all evaluated on the same 10,000 episodes. Let $\mathbf{Z}_s^A \in \{0,1\}^{40}$ record the target-exposure assignments in Arm A and $\mathbf{Z}_s^B = \mathbf{1} - \mathbf{Z}_s^A$; for target $f$, $\mathbf{Z}_{s,f}^{(1)}$ is whichever complete arm vector exposes $f$. Writing $A(m, s, f; \mathbf{z})$ for accuracy under a complete assignment, the primary estimand is the average

Table 2: Preregistered exact-exposure intervention. Accuracy is percent. Architecture intervals use Bonferroni simultaneous 99.1667% confidence; the equal-architecture macro uses the registered hierarchical 95% bootstrap.

| Architecture | Exposed | Unexposed | Effect (pp) | Simultaneous CI |
|---|---|---|---|---|
| Deep Sets | 59.60 | 53.49 | +6.110 | $[4.349, 7.849]$ |
| Relation Network | 88.34 | 87.34 | +0.996 | $[0.229, 1.749]$ |
| Set Transformer | 77.80 | 76.76 | +1.042 | $[-0.105, 2.247]$ |
| Modern Hopfield | 88.66 | 88.00 | +0.658 | $[-0.192, 1.447]$ |
| Perceiver | 74.41 | 73.12 | +1.284 | $[-0.042, 2.525]$ |
| Transformer | 82.94 | 82.04 | +0.902 | $[-0.137, 1.870]$ |
| Architecture macro | 78.63 | 76.79 | +1.832 | $[1.470, 2.189]$ (95%) |

exposed-minus-unexposed difference

$$\tau_{\text{macro}} = \frac{1}{6}\sum_{m}\frac{100}{5\cdot 40}\sum_{s,f}\left[A(m,s,f;\mathbf{Z}^{(1)}_{s,f}) - A(m,s,f;\mathbf{Z}^{(0)}_{s,f})\right]. \tag{7}$$

Because randomization flips complete curricula rather than one target at a time, cross-target transfer or interference is part of the identified contrast, not a separate quantity.

The preregistered macro criterion passes: the effect is +1.832 points, its 95% lower confidence endpoint is positive, and all six architecture estimates are positive. Only Deep Sets meets the stricter per-architecture rule (at least one point with its simultaneous interval excluding zero). Cross entropy and Brier score improve by 0.0786 and 0.0267. So exact cross-family aliases do produce a measurable endpoint effect here—small, but reliable. The design identifies a curriculum-level effect for these architectures and this budget; it says nothing about any internal memorization mechanism.

### 4.2 Study 2: A Near-Zero Contrast at High Exposure

The released Boolean in-context-learning generators (Bhattamishra et al., 2024) give the opposite outcome, and a useful counterexample to the shortcut "overlap implies a performance gain." Here the semantic IDs are exact: signed-literal vectors for Conjunction-28, active-variable masks for Majority-28, and four-element literal subsets for Sparse Disjunction-50. At the frozen 640,000-episode training budget, 0.81%, 8.88%, and 93.44% of the evaluation functions already occur in the original training streams.

Before any training, we froze 1,280 evaluation functions per task, three original streams, and paired *strict* streams that replace only the training slots colliding with an evaluation semantic ID. The released six-layer Transformer is trained for 20,000 fixed steps with three seeds under both streams. The endpoint is mean sign accuracy at prompt positions 20–39, with $\Delta_{\text{pp}} = 100(A_{\text{Original}} - A_{\text{Strict}})$.

No task reaches the frozen two-point threshold, and no task has three positive seed deltas, so neither preregistered success criterion is met and the planned seed and model expansion was not run. Even at 93.44% exact exposure, the effect stays within a tenth of a point. The result is bounded to this model, budget, and released task definitions; it does

Table 3: Boolean exposure and registered behavioral effects. Hierarchical intervals resample model seeds and semantic-ID clusters.

| Task | Exposure (%) | Mean effect (pp) | 95% CI |
|---|---|---|---|
| Conjunction | 0.81 | −0.013 | $[-0.092, 0.044]$ |
| Majority | 8.88 | −0.064 | $[-0.199, 0.078]$ |
| Sparse Disjunction | 93.44 | +0.092 | $[-0.060, 0.282]$ |

not show that exposure can never matter. Together with the small positive result above, it shows why exposure prevalence and behavioral effect are separate measurements.

## 5 Study 3: A Large Preregistered Gain on CLUTRR

The first two studies use synthetic tasks with small models; neither involves a public, natural-language-like reasoning benchmark. We therefore ran a controlled experiment on CLUTRR Sinha et al. (2019), a public benchmark of kinship reasoning from generated text. The question is narrow: *does training on one set of surface realizations of a relation pattern improve accuracy on fresh realizations of the same semantic pattern?* We report a preregistered pilot (V3) and a preregistered confirmation on disjoint targets (V4). The two studies are never pooled.

### 5.1 Task, Unit, and Design

We freeze CLUTRR Task 1 at relation length three, without distractor facts. Each example contains a three-edge kinship chain and a query relation drawn from 23 classes, and entity names are canonicalized by order of appearance. The semantic identity of an example is the ordered tuple of the target relation and the three edge relations; sentence order is a surface property and is not part of the identity.

For every three-sentence source story, we enumerate all six sentence permutations and group them under one source-group identifier. A target pattern is eligible only when it has at least 45 complete source groups. Its evaluation set contains 200 rows from the first 34 sorted groups (all six permutations of groups 0–32 and two permutations of group 33); its training exposure contains 64 rows from the next 11 groups (all six permutations of groups 34–43 and four of group 44). Evaluation and exposure therefore share no source group.

Targets are paired only within the same output relation. For each model seed, a frozen SHA-256 bit exposes one pair member in Arm A and the other in Arm B. Both arms share a 22,720-row base with no target semantics and append 1,280 target-exposure rows, giving 24,000 training rows per run with identical label counts. Initialization, all 30 epoch permutations, tensor shapes, and dropout RNG trajectories are paired across complementary runs. The model is a fixed four-layer byte-level Transformer (model dimension 128, four attention heads, feed-forward dimension 512, dropout 0.1), trained with AdamW at batch size 128 and learning rate $3 \times 10^{-4}$ under deterministic float32 CPU execution, and evaluated at the fixed epoch-30 checkpoint after 5,640 updates. There is no validation selection or early stopping, and the evaluation manifest is identical across arms within a block.

Table 4: CLUTRR exposure interventions. V3 and V4 use disjoint target semantic patterns and are not pooled.

| Study | Pairs | Targets | Runs | Effect (pp) | 95% CI; registered outcome |
|---|---|---|---|---|---|
| V3 pilot | 20 | 40 | 16 | +15.338 | [9.877, 21.369]; precision criterion not met |
| V4 confirm. | 40 | 80 | 32 | +23.339 | [17.750, 29.226]; directional criterion met |

### 5.2 Estimand and Inference

For seed $s$, let $\mathbf{Z}_s^A \in \{0,1\}^F$ record which target patterns are exposed in Arm A, with $\mathbf{Z}_s^B = \mathbf{1} - \mathbf{Z}_s^A$. For target $f$, $\mathbf{Z}_{s,f}^{(1)}$ denotes whichever complete arm assignment exposes $f$, and $\mathbf{Z}_{s,f}^{(0)}$ is the complementary assignment. Let $A_{s,f}(\mathbf{z})$ be row-weighted accuracy under complete assignment $\mathbf{z}$. For $S$ frozen seeds and $F$ frozen targets, the sole primary estimand is

$$\tau = \frac{100}{SF} \sum_{s=1}^{S} \sum_{f=1}^{F} \left[ A_{s,f}(\mathbf{Z}_{s,f}^{(1)}) - A_{s,f}(\mathbf{Z}_{s,f}^{(0)}) \right] \quad \text{percentage points.} \tag{8}$$

As in Study 1, this is the average contrast between two complete complementary curricula; it includes cross-pattern transfer rather than isolating one pattern's direct effect.

The 95% interval uses 20,000 hierarchical percentile-bootstrap replicates that resample model seeds, complete same-label target pairs, and source groups within each selected pair member, with source-group draws shared between exposed and unexposed predictions. The point estimand is the finite average over the eight frozen seeds and 80 frozen V4 targets. The interval treats the observed seeds, pairs, and within-target source groups as exchangeable draws from their empirical distributions, conditional on this model, task, eligibility filter, exposure dose, and budget; because target selection and group allocation were deterministic, it is not design-based inference for all eligible CLUTRR patterns, stories, relation lengths, or architectures. As a post-hoc sensitivity check, an intercept-only CR1 cluster-robust estimator clusters the 640 seed-pattern effects by seed and by complete target pair; the two-way cluster standard error is 2.6187 points, and a conservative $t_7$ interval is $[17.1469, 29.5313]$. This check does not replace the registered bootstrap or alter the decision rule.

### 5.3 V3: A Positive but Imprecise Pilot

V3 froze 20 same-label pattern pairs, eight model seeds, and two arms (16 runs). Its preregistration allowed a positive, negative, or equivalence conclusion only if the 95% interval was at most two points wide. The observed effect was +15.3375 points with interval $[9.8769, 21.3693]$: positive throughout, but 11.4924 points wide, so the registered outcome is "positive but imprecise." We keep that label rather than relabeling the pilot after seeing a large estimate. A post-hoc variance diagnosis attributes 94.40% of the bootstrap variance to target-pair resampling, and a calibrated projection shows that the two-point precision target would need at least 634 independent pairs while only 139 V3-disjoint eligible pairs remained. The strict-precision route was closed, and V3 served only as a pilot for designing V4.

### 5.4 V4: Preregistered Confirmation on Fresh Targets

Before any V4 training outcome existed, we froze a directional design based only on V3: a design effect of 7.66875 points (the smaller of half the V3 point estimate and its lower interval endpoint) and a 100,000-draw Monte Carlo power analysis over one to four 20-pair blocks. One block had power 0.7440 and failed; two blocks had power 0.9583 with a 95% Wilson lower bound of 0.9570 and were selected. All 80 V4 targets are semantically disjoint from the V3 targets, and the success criterion required $\tau \geq 1$ point with the two-sided 95% interval above zero.

All 32 runs reached the fixed checkpoint and passed post-training integrity checks: finite model and optimizer state, frozen epoch orders, four-way paired RNG trajectories, prediction ordering, separate-code-path metric recomputation, and artifact hashes. The confirmatory effect is +23.3391 points, 95% CI $[17.7498, 29.2257]$: the criterion is met. Both block estimates were positive (+25.0109 and +21.6672 points), as were all eight seed estimates and all 40 pair estimates; 76 of 80 target patterns were positive. These sign counts are descriptive and were not additional tests. The secondary proper losses move in the same direction (exposed-minus-unexposed cross entropy $-1.4938$, multiclass Brier score $-0.4224$, ECE-15 $-0.2018$); they were registered as descriptive and cannot alter the accuracy-based primary conclusion.

### 5.5 Why Do the Effects Differ?

The three studies span a near-zero Boolean result, a +1.8-point internal gain, and a +23.3-point CLUTRR gain. We treat this range as the main behavioral finding, not as noise to average away. A plausible reading, which we flag as post hoc: exposure buys little when the model already executes a short reusable rule (the Boolean tasks, where the released Transformer scores about 93% on Conjunction and 78–83% on Majority and Sparse Disjunction without exposure), and more where withheld compositions must be inferred from limited neighbors (CLUTRR trains a small byte-level model from scratch at one relation length and a modest budget, a regime with large seen-versus-unseen gaps). What the framework adds is that the CLUTRR gap is attributed to a certified semantic exposure difference under a preregistered, target-disjoint design rather than to a surface-form split. The implication is direct: whether semantic exposure matters is an empirical property of model, budget, and task family, so a benchmark claiming semantic novelty should measure the contrast rather than assume either a large or a negligible effect. Figure 3 places the three effects on one axis.

Within CLUTRR, pattern-level effects range from -3.9 to 75.7 pp and are strongly predicted by the unexposed baseline (Spearman $\rho = -0.847$); the median normalized gain is 0.714 (Appendix H.4). The same headroom pattern holds in the internal finite-function study (Spearman $\rho = -0.727$), where the model is near ceiling and the median normalized gain is only 0.077 (Appendix B.3). In a further exploratory (not preregistered) CLUTRR experiment, same-semantic exposure changes accuracy by 22.5 pp, versus 3.7 pp for relation-matched decoys sharing at least two edge types; the direct contrast is 18.8 pp (95% CI [13.2, 25.3]; Appendix I). The decoy contrast has a positive pattern-cluster interval, but its four-seed interval includes zero; a zero decoy effect is not excluded by the seed-level analysis.

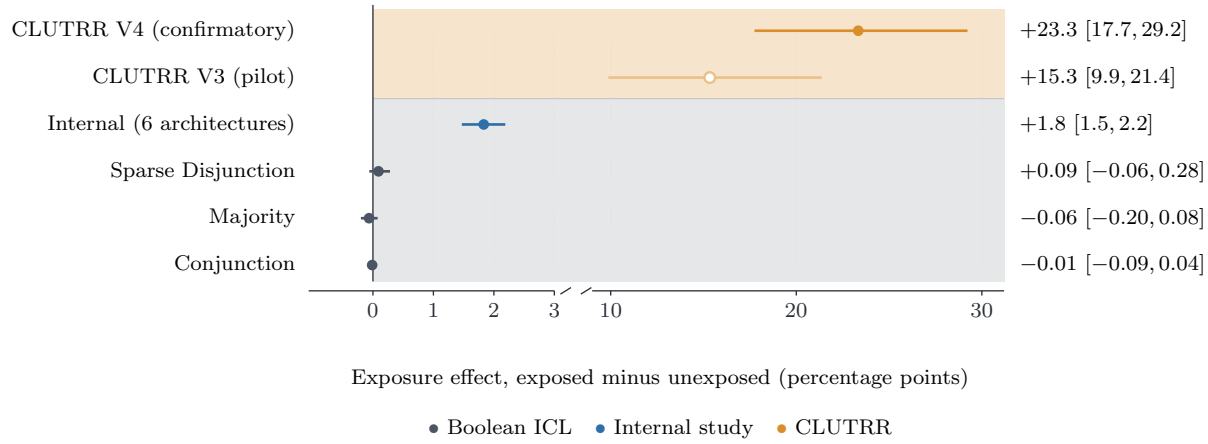



Figure 3: Exposure effects across the three controlled studies: exposed minus unexposed accuracy, in percentage points, with 95% confidence intervals. Colors group the studies: dark gray for the Boolean tasks, blue for the internal six-architecture study, orange for CLUTRR. Filled circles are confirmatory estimates; the open circle is the CLUTRR V3 pilot, which was positive but did not meet its preregistered two-point precision criterion. The horizontal axis is broken between 3 and 9 points because the Boolean and internal effects sit at zero while the CLUTRR effects are an order of magnitude larger; that gap is the finding, not a scaling artifact.

### 5.6 What the CLUTRR Result Does and Does Not Say

The experiment identifies an endpoint effect of randomized complementary curricula: the curriculum that exposes a focal relation composition yields higher focal-pattern accuracy on different source groups and sentence permutations, a behavioral consequence beyond exact-string reuse that the structural audits lack. It does not isolate one pattern's direct effect from transfer among the other patterns whose assignments also reverse, and the claim stays bounded to one task, relation length, architecture, exposure dose, and budget. It does not show that the model stores a literal lookup table, or that the effect holds for unseen lengths, family-disjoint reasoning, other architectures, or natural text. The near-zero Boolean result remains a counterexample to "overlap implies memorization," so the combined evidence supports the weaker conclusion: curricula that differ in semantic exposure can have a large behavioral effect, and benchmark novelty and behavioral exploitation must be measured separately.

## 6 Auditing Three Public Protocols

The controlled studies estimate behavior when retraining is feasible. The structural audits here answer the prior question: does a released protocol support the semantic novelty it

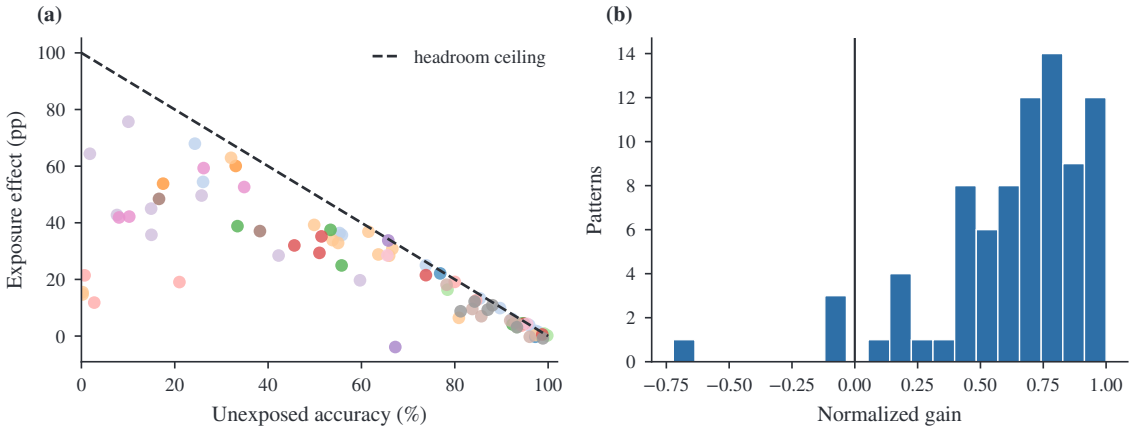



Figure 4: Exploratory (not preregistered) heterogeneity across the 80 CLUTRR V4 semantic patterns. In (a), points are colored by target relation and the dashed line marks the headroom ceiling; effects are largest when unexposed accuracy leaves more headroom. Normalized gain (b) remains high for most patterns.

claims? Each case keeps its own semantic namespace, witness completeness, and claim boundary; we do not pool them into a prevalence estimate.

### 6.1 SLR-Bench: String Novelty Masks Logical Overlap

SLR-Bench contains 20 curriculum levels generated by an automated logical-reasoning framework (Helff et al., 2026); its released `v1-All` split has 18,053 training rows and 1,000 test rows, and no test rule string appears verbatim in training. String exclusion, however, is not logical novelty. Tokenizing Prolog while ignoring whitespace already finds 70 test rules identical to a training rule. For flat positive rules, we additionally canonicalize variable names and condition order, drop duplicate existential profiles, and drop existential conditions that a stronger profile already entails. This proves 382 further train–test equivalences. The two sets are disjoint, so

$$\frac{70 + 382}{1000} = 45.2\% \tag{9}$$

is a sound lower bound (Figure 1).

The mechanism is ordinary logic, not fuzzy similarity. Without variable inequality,

$$\exists x\, P(x) \wedge \exists y\, P(y) \equiv \exists x\, P(x), \tag{10}$$

because both variables may bind to the same object. Likewise, a weaker existential condition is redundant when a stronger profile entails it. Every reported pair comes with a train rule, a test rule, and the canonical witness, so the 45.2% figure can be checked pair by pair rather than believed.

We do not infer model impact from the released per-item prediction artifact: although it aligns with all 1,000 test IDs and reference rules, its Basic-tier accuracy does not reproduce the paper's reported aggregate. A preregistered adjusted overlap–correctness analysis

also stopped before fitting, because the output is not tied to a reproducible checkpoint and exact curriculum-level common support retains only 352/452 witnessed rows (77.88%), below the frozen 80% threshold. SLR therefore supports the structural lower bound, not a performance-inflation estimate.

For the 548 unresolved SLR-Bench items, an exploratory (not preregistered) sensitivity analysis leaves the correct fraction $p$ unspecified: Proposition 2 gives $[\max(0, pM - U), \min(pM, M-U)]/(M-U)$ for an assumed exposure upper bound $U < M$. Appendix J.3 shows how wide the interval can get, without using the released prediction artifact or estimating model performance.

### 6.2 Two More Failure Modes: Broken Witnesses and Resampled Pools

**Witness breakage (Tensor Logic).** The compositional split of Shah and Zadrozny (2026) selects two-hop test paths and then removes the 1,000 direct test edges from training. Edge removal works; path removal does not. Because the removal is global, it can delete an edge that another selected path relies on, and only 379 of the 1,000 designated test paths survive. Allowing alternative intermediate entities, 484 of 1,000 queries retain an endpoint-level witness in training, while the ordered relation-composition schema appears in scrubbed training for 999 of 1,000. Equation 6 therefore gives endpoint exposure $[484, 1000]/1000$ and schema exposure $[999, 1000]/1000$. The repository publishes no checkpoint, per-query predictions, or training logs, so no behavioral effect is identified.

**Finite-space resampling (Occam).** The released Markov evaluator of Deora et al. (2025) fixes ten task seeds per order, while training assigns seed `step+i`. Under the paper's 800,000-sequence budget, evaluation seeds 5 and 12,579 occur in training six and 32 times, respectively. The assignment of orders uses unseeded NumPy state and was not persisted, so we cannot tell which order each occurrence used; grouping the two evaluated orders per seed gives a certified interval of $[2, 4]/20$ exact transition-task exposures in each main experiment, or 10–20%. An exhaustive census of 75,093 potential training transition signatures finds no cross-seed collision, and at least 93.7425% of training episodes reuse a prior `(order, transition tensor)` because adjacent batches reuse seeds. The pinned repository contains no linear-regression implementation, although regression was part of the frozen audit scope; that scope remains unresolved rather than replaced by the available Markov subset.

### 6.3 Reporting Order and the Failed Breadth Criterion

Tensor Logic and Occam were selected and frozen before we looked at any semantics, and both are reported. Our original breadth criterion—two complete public audits and one credible behavioral source—was not met, and we keep that outcome instead of redefining the criterion: the separately preregistered CLUTRR study supplies behavioral evidence without changing the historical decision. Appendix F gives the full registration timeline. These cases are protocol audits, not prevalence estimates.

## 7 From Audits to a Reporting Standard

The three failure modes are cheap to prevent once they are named. We distill the audits into a short reporting standard for any benchmark that claims novelty, and into a toolkit that checks the standard mechanically.

1. *Name the held-out object.* State whether the claim concerns records, parameters, programs, complete functions, relation compositions, or rule families, together with the observation contract, and pass the must-match/must-differ adequacy test before measuring overlap. "Unseen" without a named object and its observable equivalence is not a specification.

2. *Report sensitivity to the unit.* Run the audit under every unit that passes the adequacy test and report whether conclusions change; endpoint paths and relation schemas are both valid units for Tensor Logic, with very different answers.

3. *Split after semantic grouping.* When exact semantics are finite, canonicalize or hash complete mappings before splitting; for executable programs, use proof-producing normalization under the contract, and label incomplete detection with Equation 6 rather than zero overlap.

4. *Record feasibility before splitting.* Specify the semantic domain, the basis for equivalence, normalization and witness-search costs, and a stopping rule fixed independently of observed overlap (Appendix A.3); exhaustive procedures support exact grouping, sound but incomplete ones support a lower bound, and approximate ones need an error model.

5. *Count the training budget.* Report weighted coverage using Equation 5, not the nominal hypothesis-space size; online generation does not guarantee semantic novelty.

6. *Re-check witnesses after mutation.* Removing one edge, rule, or source object can invalidate another selected witness, so rerun semantic validity checks on the final materialized data.

7. *Measure behavior separately.* Prevalence and impact are different quantities: per-item predictions permit descriptive stratification, while stronger claims require a matched intervention.

8. *Keep provenance.* Manifests should record contracts, semantic IDs, source hashes, seeds, category assignments, witness grades, validator coverage, and residual risk; tables should be generated from structured results, and negative outcomes should not be rewritten after a later success.

**The toolkit.** The versioned `semantic-audit-v1` command maps a benchmark into a claim-specific schema recording the fields above: observation contract, semantic units, source hashes, witnesses, completeness, bounds, and permitted claims. Validation rejects unsupported overlap decisions and missing contracts; canaries, negative tests, and two isolated reproductions pass with byte-identical manifests (Appendix K). The supplement retains frozen manifests, paired-run checks, predictions, bootstrap draws, and separate-code-path metric recomputation, and `make reproduce-paper` regenerates the analyses, macros,

Table 5: Positioning against the nearest contamination and audit work. *Object*: what counts as a duplicate. *Evidence*: the strongest guarantee attached to a finding. *Behavioral*: whether a behavioral effect of exposure is measured.

| Work | Object | Evidence | Behavioral |
|---|---|---|---|
| Surface detectors (Sainz et al., 2023; Xu et al., 2024; Golchin and Surdeanu, 2024; Oren et al., 2024) | surface text of fixed test sets | detector estimate | none |
| Deduplication studies (Lee et al., 2022; Allamanis, 2019) | near-duplicate documents | statistical estimate | indirect |
| Magar and Schwartz (2022) | downstream examples in pretraining | exact (surface) | controlled |
| Spiesberger et al. (2026) | semantic duplicates | similarity estimate | controlled |
| This paper | generator semantics under a contract | certified: exact, sound lower bound, or interval | preregistered, paired curricula |

and figures from compact results without retraining; Appendix L maps claims to artifacts. The artifact is available at `https://github.com/zihanzhou0725/semantic-novelty-audit-artifact`. The shared schema preserves distinct semantic units rather than imposing a pooled metric.

## 8 Related Work

Table 5 positions this paper against the nearest lines of work; the text below expands on each.

**In-context learning and procedural generalization.** Studies of in-context function learning (Garg et al., 2022; Akyürek et al., 2023; Bhattamishra et al., 2024), meta-learning (Finn et al., 2017; Vinyals et al., 2016), and systematic generalization (Lake and Baroni, 2018; Chollet, 2019) ask what algorithms or abstractions models acquire. We ask a prior question: whether the evaluation protocol has isolated the semantic object that such a claim depends on.

**Contamination and semantic identity.** Leakage can invalidate conclusions even when performance is measured correctly (Kapoor and Narayanan, 2023). Most contamination work operates on the surface text of fixed test sets (Sainz et al., 2023; Xu et al., 2024; Golchin and Surdeanu, 2024; Oren et al., 2024), and deduplication studies show that near-duplicates inflate measured generalization (Lee et al., 2022; Allamanis, 2019). Our objects are different: generator semantics that can recur under arbitrarily different surface forms. The nearest comparator is Spiesberger et al. (2026), who detect semantic duplicates in opaque corpora and test models trained on synthetic semantic variants. Our increment is narrower and executable: a contract defines equality, exact or proof-producing signatures

replace similarity where available, and unresolved identities stay explicit instead of being forced into a detector decision.

**Behavioral interventions.** Magar and Schwartz (2022) separate memorization from exploitation by manipulating whether downstream examples appear in pretraining, and Spiesberger et al. (2026) fine-tune on semantic duplicates. Our design randomizes exact semantic-pattern assignments in paired complementary curricula, keeps target-level prediction manifests, and treats structural exposure, the curriculum-level contrast, and internal mechanisms as three separate quantities.

**Generated reasoning benchmarks.** CLUTRR (Sinha et al., 2019), SLR (Helff et al., 2026), Tensor Logic (Shah and Zadrozny, 2026), and the Occam experiments (Deora et al., 2025) supply the protocols we audit; we do not compare model rankings across them. Dynamic benchmarks such as DyVal (Zhu et al., 2024) and Reasoning Gym (Stojanovski et al., 2025) regenerate instances to resist contamination; our Occam audit shows that regeneration alone does not guarantee semantic novelty when the hidden task space is finite.

**Compositional-generalization splits.** SCAN (Lake and Baroni, 2018), COGS (Kim and Linzen, 2020), and the maximum-compound-divergence splits of CFQ (Keysers et al., 2020) build held-out sets from a known grammar so that test items require unseen combinations of seen primitives; Hupkes et al. (2020) decompose the corresponding notions of compositionality. These works design new splits with semantic control built in; we go the other way and audit existing protocols whose splits are defined on representations. The two views are complementary: compound divergence could itself serve as a semantic unit inside an observation contract.

**Partial identification.** The certified interval follows the partial-identification principle that assumptions and verified constraints determine a set of compatible values, not necessarily a point (Tamer, 2010). The nearest analogue is Balke and Pearl (1997), whose linear program bounds treatment effects under imperfect compliance; we claim no new partial-identification theory. Our specialization uses disjoint certificate groups to get a closed-form finite identified set, ties every constraint to row-level provenance, reports machinery assurance separately, and keeps behavioral effects as separate estimands.

**Evidence schemas.** Datasheets (Gebru et al., 2021) and Croissant (Akhtar et al., 2024) document dataset provenance and structure. Our schema is complementary and adds the fields specific to semantic-novelty evidence: the contract and unit, witness grade and completeness, certificate constraints and bounds, behavioral-estimand metadata, residual risk, and the allowed claim wording.

## 9 Limitations

Our settings are synthetic, and deliberately so: they are the cases where semantic identity can be checked exactly or with proofs. Whether natural language benchmarks have similar overlap rates is a different question, which our audits do not answer. Within our settings, several boundaries deserve emphasis.

The SLR result is a sound lower bound: complex aggregates, negation, arithmetic, and unresolved disjunction are not fully normalized, so the true overlap is at least 45.2% and plausibly higher. The released SLR model output does not reproduce the reported aggregate, so we make no behavioral claim for SLR. Tensor Logic and Occam lack the artifacts needed for a behavioral estimate, and one frozen Occam scope (linear regression) is unimplemented in the pinned repository.

Our contracts cover finite, deterministic, effect-free tasks. We have not built equivalence procedures for partial, stateful, nondeterministic, interactive, exception-raising, or resource-sensitive programs. The certified intervals are conditional on the audited sources and the semantic-signature machinery; the qualitative assurance risks in Appendix K are not converted into a numerical error rate.

Each intervention covers a bounded population. The internal study spans six architectures, 40 cross-family targets, and one budget, and only Deep Sets passes the per-architecture simultaneous criterion; the Boolean null covers one released Transformer configuration and three task definitions, and with three seeds between-seed uncertainty is coarse, so hierarchical and Student-$t$ intervals are both reported; CLUTRR V4 covers one task, relation length, architecture, exposure dose, and budget, so its large effect need not extend to other lengths, architectures, or natural text. Both positive studies randomize complete curricula, so the contrast includes cross-target transfer rather than isolating a single target. The CLUTRR V3 pilot did not meet its preregistered precision criterion despite a positive interval, so it is not pooled with V4; the certified interval is tight only relative to the supplied certificate system, and no study identifies a literal lookup table or other internal mechanism.

## 10 Conclusion

Syntactic novelty is not semantic novelty. Across six generated reasoning settings, representation-level holdouts admitted exact aliases, provable logical equivalences, broken executable witnesses, or repeated hidden tasks. Certified intervals keep incomplete witness search honest instead of turning missing evidence into novelty. And the behavioral question has no default answer: paired curricula with exact semantic exposure produced a +1.832-point macro gain across six architectures and a +23.339-point gain on a public benchmark, while a Boolean task moved by less than a tenth of a point at 93.44% exposure. Benchmarks should name the object their split holds out, grade their evidence, and report the interval; and anyone who wants to know whether exposure matters should measure it.

## Broader Impact Statement

Our work audits public benchmarks and gives benchmark authors a reporting standard and a checking toolkit. The intended impact is a more trustworthy evaluation ecosystem: novelty claims become checkable, and our per-item witnesses let anyone verify the audits instead of trusting them. We see two risks. First, a result like our 45.2% lower bound for SLR-Bench could be misread as "the benchmark is useless." We do not claim that: overlap prevalence is not a performance-inflation estimate, and our own behavioral studies show

that exposure can have a near-zero effect. We ask readers to report our bounds together with their evidence grades. Second, auditing named protocols can read as blame. We treat the audited systems as evidence of a general measurement gap, not as faults of their authors, whose publicly released tooling made our analysis possible; every finding is released with per-item witnesses so that it can be checked and disputed on the evidence rather than on our summary. All data we use are synthetic or public, and we see no privacy or security concerns.

## Data and Code Availability

All audited artifacts are public and were pinned by revision and SHA-256 hash: SLR-Bench `v1-All` and its model outputs (Helff et al., 2026), the Boolean in-context-learning generators (Bhattamishra et al., 2024), CLUTRR (Sinha et al., 2019), Tensor Logic (Shah and Zadrozny, 2026), and the Occam repository (Deora et al., 2025). Our code, frozen protocols, manifests, per-item predictions, and analysis results are available at `https://github.com/zihanzhou0725/semantic-novelty-audit-artifact`, and `make reproduce-paper` rebuilds every table, macro, and figure in this paper from the retained results without retraining. Training checkpoints (1.9 GB) are excluded and are not needed to reproduce any reported statistic.

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## Appendix A. Audit Definitions and Evidence Grades

### A.1 Evidence Grades

Table 6: Semantic-equivalence evidence grades.

| Grade | Meaning |
|---|---|
| Exact | Equality of complete finite maps or a task-specific injective semantic ID. |
| Sound lower bound | Every detected pair is provably equivalent, but unresolved objects may add collisions. |
| Estimate | Equality is inferred from incomplete execution or statistical signatures and requires an error analysis. |

The internal benchmark and selected Boolean classes use exact signatures. SLR Helff et al. (2026) uses token identity plus proof-producing normalization for a flat positive fragment, so 45.2% is a sound lower bound.

### A.2 Audit Procedure

For each benchmark we follow a fixed procedure: declare the representation and family objects, observation contract, and advertised claim; freeze source and dataset revisions plus a must-match/must-differ falsification set; construct and validate semantic signatures, rejecting units that split must-match pairs or merge must-differ pairs; measure train–test overlap or weighted training-time coverage; preserve row-level witnesses or function manifests; if behavior is claimed, compare paired per-item outputs or run a matched intervention; label every result exact, lower-bound, or estimated and record implementation assurance separately; and, when several units survive, report the audit under each and flag any conclusion that changes.

For a discrete function distribution $p(f)$, the exact probability that an independent test function was sampled at least once in $N$ training draws is

$$P_{\text{seen}}(N) = \sum_f p(f) \left[1 - (1 - p(f))^N\right]. \tag{11}$$

### A.3 Feasibility of Semantic Normalization

Before constructing a split, benchmark designers should freeze the specifications in a feasibility record and report the residual unresolved count or mass when the stopping rule is reached. The record contains:

1. the observation contract and claim-relevant semantic unit;
2. whether equivalence is decidable, or the exact bounded domain on which it will be analyzed;

3. the expected normalization cost per object and at dataset scale;
4. the witness-search budget in time, states, or candidate aliases;
5. a stopping rule fixed independently of observed overlap;
6. the argument that every emitted certificate is sound; and
7. the residual unresolved count or mass at the stopping rule.

## Appendix B. Internal Finite-Function Audit

### B.1 Task and Semantic Identity

The domain and codomain are $\mathcal{X} = \mathcal{Y} = \{0, \ldots, 19\}$. Each episode samples a rule, reveals $K = 8$ distinct support sources with independent label corruption probability $\rho = 0.1$, and queries a clean target at a source outside the support set. Complete semantic identity is the 20-output vector

$$q(r) = [f_r(0), \ldots, f_r(19)]. \tag{12}$$

The three training families contain 560 parameter identities and 518 unique complete functions.

The legacy split shuffles parameter identities, holds out 140, and samples identities uniformly. In the five matched public-model seeds, the number of held-out identities whose function appears in training is $13, 13, 19, 15, 18$ (mean 15.6/140, or 11.14%). The corrected split groups identities by $q(r)$, stratifies canonical functions by family, and samples functions uniformly. It contains 388 training and 130 test functions with zero complete-map overlap.

### B.2 Complementary Exact-Exposure Intervention

The confirmatory intervention targets all 40 functions with identities in multiple registered families. Every target has an affine evaluation identity and a parity training alias; 38 have signature `affine+parity`, while two also have a block identity. For every intercept $b \in \{0, \ldots, 19\}$, we pair affine functions $f_{1,b}$ and $f_{11,b}$. A frozen SHA-256 bit assigns one pair member to Arm A and the other to Arm B for each model seed. Hence every target is exposed in exactly one arm and unexposed in the complementary arm.

Each arm contains 478 common base functions and 20 exposed target functions. The paired 20,000-episode streams share base slots, support/query draws, corruption, initialization, epoch permutations, optimizer schedule, and the 10,000-episode evaluation manifest. Six architectures, five seeds, and two arms give 60 runs. Every run reaches the fixed epoch-60 checkpoint after 4,740 optimizer updates; no validation selection is used.

For seed $s$, let $\mathbf{Z}_s^A$ be the complete 40-target exposure assignment in Arm A and let $\mathbf{Z}_s^B = \mathbf{1} - \mathbf{Z}_s^A$. For target $f$, $\mathbf{Z}_{s,f}^{(1)}$ denotes whichever arm assignment exposes $f$, and $\mathbf{Z}_{s,f}^{(0)}$ denotes its complement. For architecture $m$, let $A(m, s, f; \mathbf{z})$ be mean accuracy under

complete assignment $\mathbf{z}$. The sole primary estimand is

$$\tau_m = \frac{100}{5 \cdot 40} \sum_s \sum_f \left[ A(m, s, f; \mathbf{Z}^{(1)}_{s,f}) - A(m, s, f; \mathbf{Z}^{(0)}_{s,f}) \right], \tag{13}$$

$$\tau_{\text{macro}} = \frac{1}{6} \sum_m \tau_m. \tag{14}$$

The contrast randomizes complete complementary curricula, not a focal target's exposure while holding all other targets fixed. The 95% macro interval uses 20,000 hierarchical bootstrap replicates over model seeds, 20 matched function pairs, and evaluation episodes within function; architectures retain equal fixed weights. Model intervals use 99.1667% Bonferroni simultaneous confidence.

All six architecture point estimates are positive; only Deep Sets and Relation Network have simultaneous 99.17% intervals that exclude zero, and the per-architecture 95% intervals appear in Table 2. Cross entropy changes from 0.91052 to 0.83191 (exposed minus unexposed $= -0.07861$), and Brier score changes from 0.34501 to 0.31829 ($-0.02673$). Training probabilities were not persisted, so registered training calibration is not estimable and is not replaced by loss.

### B.3 Exploratory Function-Level Heterogeneity

An exploratory (not preregistered) function-level decomposition finds positive effects for all 40/40 target functions, with range 0.13–4.48 pp and median 1.80 pp. As in CLUTRR, the effect is negatively associated with unexposed accuracy (Spearman $\rho = -0.727$). The median normalized gain is only 0.077: the model is close to ceiling and exposure moves only a small fraction of the remaining headroom.

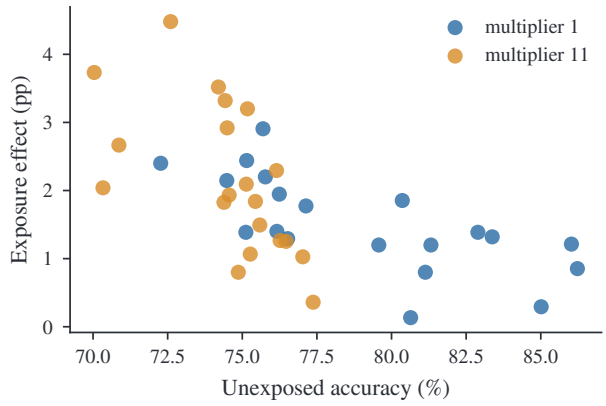



Figure 5: Exploratory internal function effects against unexposed accuracy. Color distinguishes the two affine multipliers.

The normalized-gain contrast is substantial: exposure removes much less of the remaining error in the internal setting than in CLUTRR (Spearman $\rho = -0.727$ versus $-0.847$;

median normalized gain 0.077 versus 0.714). This analysis does not evaluate displacement on non-target functions.

## Appendix C. SLR-Bench Semantic Audit

### C.1 Frozen Sources

Table 7: Frozen SLR audit inputs. Individual Parquet SHA-256 values are stored in the artifact manifest.

| Object | Frozen value |
|---|---|
| Repository | `ScalableLogicalReasoning` |
| Code revision | `3b46979ccdf9...` |
| Dataset | `AIML-TUDA/SLR-Bench`, configuration `v1-All` |
| Dataset revision | `cecc0aa26029...` |
| Rows | 18,053 train; 1,000 test |

### C.2 Token Identity

After comments are removed, the Prolog lexer preserves atoms, variables, punctuation, and operators while discarding whitespace. Seventy large language model (LLM)-guided test rules have a token sequence equal to a training rule. The raw rule strings remain different, so verbatim overlap is zero.

```
% test
eastbound(T) :-
  has_car(T, C),
  (has_wall(C, full);
   car_color(C, yellow)).

% train: same Prolog tokens
eastbound(T) :-
  has_car(T, C),
  (has_wall(C, full);
   car_color(C, yellow)).
```

### C.3 Flat Positive Rule Semantics

A parsed flat positive rule has the form

$$\text{eastbound}(t) \leftarrow \bigwedge_j A_j(t, \mathbf{z}). \tag{15}$$

For each existential car variable, we collect its unary attribute profile, sort atoms within the profile, sort profiles, and alpha-rename variables. Three successively stronger canonicalizations are used:

1. alpha-renaming and conjunction-order normalization;
2. idempotence of duplicate existential profiles;
3. removal of a profile entailed by a stronger profile.

Idempotence is sound because variables need not denote distinct objects:

$$\exists x\, P(x) \wedge \exists y\, P(y) \equiv \exists x\, P(x). \tag{16}$$

For profile sets $P \subseteq Q$,

$$\exists x\, Q(x) \wedge \exists y\, P(y) \equiv \exists x\, Q(x), \tag{17}$$

because an object satisfying $Q$ also witnesses $P$.

Table 8: Progressive SLR overlap counts out of 1,000 test rows. The flat-rule parser covers 702 test and 13,568 training rows.

| Canonicalization | Overlapping test rows |
|---|---|
| Alpha-renaming + atom order | 81 |
| + existential idempotence | 369 |
| + profile subsumption | 382 |
| Token identity (disjoint set) | 70 |
| Sound union lower bound | **452** |

The union contains 382 random-generator rules and 70 LLM-guided rules, with zero intersection. It covers every test row at curriculum levels 1 and 3, and remains nonzero through level 20. Aggregate, negated, arithmetic, and non-token-identical disjunctive rules are unresolved, so 45.2% cannot be interpreted as an exact total.

### C.4 Released Per-Item Output Caveat

The repository `AIML-TUDA/Llama-3.1-8B-SLR` contains 1,000 per-item outputs whose problem IDs and ground-truth rules align exactly with `v1-All`. A verifier mirroring the released extensional judge, run with Trealla Prolog, gives tier accuracies of 51.6/55.2/20.4/5.2 (Basic/Easy/Medium/Hard) for the hosted artifact, against 96/54/18/5 in the ACL 2026 Table 2 and 92/77/17/2 in arXiv v2. Even treating all 30 Basic-tier runtime errors as correct yields only 63.6%, so runtime differences cannot explain the final-paper gap.

We therefore mark this artifact `exploratory_only` and do not use it to estimate the reported model's seen/unseen effect.

### C.5 Registered Association: Not Identified

The confirmatory target was the final-paper `Llama-3.1-8B-IT-SLR` result. Before loading outcomes, the protocol required a prediction source tied to a checkpoint and complete decoding configuration, plus at least 80% retention in both witnessed and no-witness groups under exact curriculum-level common support.

No eligible prediction source was found. The repository-hosted rows align with all 1,000 test IDs and rules but do not reproduce the published main table. The common-support precheck also retains only 352/452 witnessed rows (77.88%), versus 498/548 no-witness rows (90.88%). The registered propensity model, outcome model, and 10,000-replicate bootstrap were therefore not run. No adjusted association estimate, confidence interval, or direction is reported.

## Appendix D. Boolean ICL Generator and Coverage Audit

### D.1 Released Version

The camera-ready paper Bhattamishra et al. (2024) points to `satwik77/incontext-bool`. We freeze revision `a25a2cb0c62...`; its `bool_tasks.py` SHA-256 is `5bda5cdb00bc...`. Full hashes are frozen in the source manifest. The core task abstract syntax trees (ASTs) match the camera-ready supplementary archive. No checkpoint or per-function evaluation output is released.

Table 9: Paper/released-code differences. The behavioral intervention uses only the frozen released definitions and is not presented as a reproduction of unreleased main-table configurations.

| Task | Camera-ready description | Released code |
|---|---|---|
| Sparse Disjunction | $n = 50$, at most three signed literals | Exactly four literals, positive only |
| Sparse Parity | Main table includes $(10, 2)$ and $(20, 3)$ | Sampler hard-codes $k = 2$ and ignores a $k$ argument |
| 0–1 Threshold | Uniform weights and seven integer thresholds | Weights use $(0.7, 0.15, 0.15)$ and six half-integer thresholds |
| Registry | 0–1 Threshold / Integer Halfspace task labels | Names are `sparse_thres` / `int_halfspace`; bundled alternative names are unregistered |

### D.2 Exact Semantic IDs

For Conjunction-28, the semantic ID is the vector $w \in \{0, +1, -1\}^{28}$; exhaustive small-dimension truth tables verify injectivity, and differing signed literal sets can be separated by an input. For Majority-28, the ID is the active-variable bit mask. For released Sparse Disjunction-50, the ID is the four-element positive-literal subset. Any two different subsets are separated by assigning a differing literal true and the other subset's literals false. Thus the IDs are exact for these released families without enumerating $2^{28}$ or $2^{50}$ inputs.

At the frozen budget $N = 20{,}000 \times 32 = 640{,}000$, the analytic exposure probabilities from Equation 11 are 0.914%, 9.457%, and 93.790%. The actual frozen streams match these closely: 0.81%, 8.88%, and 93.44% of the 1,280 frozen evaluation draws (per-seed variation at most 0.4 points), and every Strict value is zero.

## Appendix E. Boolean Behavioral Intervention

### E.1 Preregistration and Pairing

Before training we froze 1,280 evaluation function draws and prompt seeds per task, three 640,000-function Original streams, Strict streams that replace only slots colliding with any evaluation semantic ID, and all model seeds, input random-number streams, optimizer, budget, primary endpoint, bootstrap procedures, and success criteria.

The model is the released Transformer implementation configured with six layers, hidden size 256, eight heads, no dropout, and float32 precision. Training uses Adam with learning rate $10^{-4}$, batch size 32, 40 points per episode, and 20,000 fixed steps. No validation selection is used. The final checkpoint is evaluated on the same prompts under both protocols.

The primary endpoint is mean sign accuracy over zero-indexed positions 20–39. The effect is

$$\Delta_{\text{pp}} = 100(A_{\text{Original}} - A_{\text{Strict}}). \tag{18}$$

The core criterion requires one task with mean $\Delta_{\text{pp}} \geq 2$ and all three seed deltas positive. The strong-pattern criterion additionally requires Sparse Disjunction $\geq 5$ points, absolute Conjunction effect $\leq 2$ points, and nondecreasing effects across the exposure gradient. Confidence intervals are reported but do not determine the decision.

### E.2 Complete Primary Results

Table 10: Per-seed primary results. Each interval uses 10,000 paired semantic-ID cluster bootstrap replicates.

| Task | Seed | Exposure (%) | Original (%) | Strict (%) | Δ (pp) | Cluster-bootstrap 95% CI |
|---|---|---|---|---|---|---|
| Conjunction | 0 | 0.94 | 92.871 | 92.875 | -0.004 | $[-0.094, 0.090]$ |
| | 1 | 0.70 | 93.059 | 93.117 | -0.059 | $[-0.203, 0.086]$ |
| | 2 | 0.78 | 92.902 | 92.879 | +0.023 | $[-0.004, 0.051]$ |
| Majority | 0 | 8.83 | 78.379 | 78.520 | -0.141 | $[-0.324, 0.039]$ |
| | 1 | 8.28 | 78.852 | 78.953 | -0.102 | $[-0.281, 0.078]$ |
| | 2 | 9.53 | 79.223 | 79.172 | +0.051 | $[-0.129, 0.230]$ |
| Sparse Disjunction | 0 | 93.05 | 81.387 | 81.418 | -0.031 | $[-0.180, 0.125]$ |
| | 1 | 92.81 | 83.613 | 83.367 | +0.246 | $[-0.023, 0.516]$ |
| | 2 | 94.45 | 81.949 | 81.887 | +0.063 | $[-0.078, 0.203]$ |

No task reaches 2 points or has three positive deltas. Neither the core nor the strong-pattern criterion is met. A $t_2$ interval is wider than the hierarchical one but also includes zero for every task. The registered Transformer seed expansion and long short-term memory (LSTM) ranking experiment are not triggered.

### E.3 Secondary Endpoints

Secondary means (Original minus Strict) are uniformly small: final-position accuracy +0.05, −0.13, and +0.16 pp; positions 1–39 +0.01, −0.07, and +0.07 pp; and tail cross-entropy +0.0007, −0.0001, and −0.0019 for Conjunction, Majority, and Sparse Disjunction. These endpoints do not replace the registered primary criterion.

Within Original runs, seen-versus-unseen accuracy is descriptive only: exposure is not randomized and the groups differ substantially in size. We do not use this stratification as a substitute for the paired intervention.

### E.4 Integrity Validation

All 18 runs completed 20,000 steps. A separate validation code path checks:

- frozen source, adapter, manifest, and stream hashes;
- finite model and optimizer tensors;
- 1,280 finite per-episode logits per run;
- zero Strict semantic exposure;
- identical paired evaluation semantic IDs, labels, and Original exposure flags;
- complete finite training logs and artifact SHA-256 values.

All checks pass. The compact analysis contains 11,520 paired episode rows. The 1.9 GB checkpoints are not required to reproduce the reported statistics and are excluded from the public artifact.

## Appendix F. Registration Timeline

Rows are listed in evidential order and are not pooled as one sequential experiment. Each row states what was frozen before the corresponding outcomes were observed and which earlier decision later work does not revise.

Table 11: Registration timeline. Each study was frozen before its outcomes were observed; later studies do not revise earlier decisions.

| Study | Frozen before outcomes | Role and outcome | Not revised by later work |
|---|---|---|---|
| Internal intervention | Paired streams, endpoints, and success criteria before training | Confirmatory exact-alias intervention; preregistered macro criterion met | Later public studies do not alter it |
| SLR-Bench | Source artifacts and identification rules before outcome modeling | Structural 45.2% lower bound; association analysis stopped at its preregistered identification check | No later outcome substitutes for the stopped analysis |
| Original breadth scope (Tensor Logic, Occam) | Candidates before semantic extraction | Prospective structural audits; breadth criterion not met | CLUTRR uses a new scope and does not reopen it |
| Toolkit and theory validation | Schema tests and certificate assumptions before validation runs | Validate reporting machinery | Do not change the breadth decision |
| CLUTRR V3 | Targets, training, endpoint, bootstrap, and precision criterion | Preregistered pilot; positive but imprecise | The positive direction is not relabeled as success |
| V3 diagnosis | Exploratory status declared after V3 was closed | Post-unblinding variance analysis | Designs V4 but cannot change V3 |
| CLUTRR V4 | V3-disjoint targets and directional criterion before materialization and training | Prospective confirmation; directional criterion met | Not pooled with V3 |

## Appendix G. Original Prospective Public-Benchmark Scope

Candidate discovery queried Semantic Scholar and arXiv using four frozen strings, deduplicated by DOI, arXiv ID, then normalized title, and screened in ascending title order. The manifest was frozen before candidate code or data were semantically inspected. Both selected candidates are reported; neither was removed after its result was known.

### G.1 Tensor Logic FB15k-237

We pin the Tensor Logic code at revision `9b497c279928...` and FB15k-237 at `2e440e0f9c68...`. Experiment 3B first selects 2,000 unique endpoint pairs with two-hop paths, splits them equally into validation and test, and then removes all corresponding direct triples from training. The code does not revalidate whether removing one direct triple destroys an edge used by a different selected path.

Table 12: Tensor Logic path audit, out of 1,000 rows per split. Endpoint and relation-schema witnesses answer different semantic questions.

| Audit quantity | Validation | Test |
|---|---|---|
| Direct edge absent after scrub | 1,000 | 1,000 |
| Designated two-hop path retained | 336 | 379 |
| Any exact endpoint witness retained | 441 | 484 |
| Relation schema seen in scrubbed train | 994 | 999 |
| No retained endpoint witness | 559 | 516 |

Thus direct-edge withholding succeeds, but the claim that every evaluation query retains its generating two-hop path fails. Under the certified interval below, the test split has endpoint exposure in $[484, 1000]/1000$ and relation-schema exposure in $[999, 1000]/1000$. No public checkpoint, per-item ranks/predictions, or training logs were found, so this is a protocol-level audit without credible behavioral evidence.

### G.2 In-Context Occam's Razor

We pin the public repository at revision `3b8fe0d69dd9...`. Its Markov training generator assigns task seed `step+i`; the evaluator fixes ten seeds $s_j = 12574j + 5$ for every tested order and draws 100 sequences per task. The paper's 800,000-sequence horizontal axis and batch size 32 imply 25,000 nominal steps, hence training seed support $0, \ldots, 25030$.

Seeds 5 and 12,579 occur six and 32 times, respectively. Because the category assignment chooses between the two trained orders but uses unseeded NumPy state, the exact exposed order identities are unavailable. For each supported seed, however, at least one and at most both evaluated orders occur in training. An exhaustive census recomputes 75,093 transition signatures (25,031 seeds for orders 1, 2, and 3) and finds no cross-seed collision. Consequently both main experiments have exact latent-task exposure $[2, 4]/20 = [10\%, 20\%]$, corresponding to 200–400 of 2,000 evaluation sequences.

The same seed rule creates 800,000 training episodes from only 25,031 seed IDs. Even if every seed realizes both orders whenever possible, there are at most 50,060 distinct `(order, transition tensor)` tasks, so at least 749,940/800,000 (93.7425%) episodes reuse a latent task. The repository has no linear-regression generator, training, evaluation, or curriculum code, although regression is part of the frozen candidate scope. That scope is therefore unresolved and the candidate is not a complete audit under the original breadth scope.

### G.3 Frozen Breadth Decision

Tensor Logic counts as one complete public semantic audit; the Occam candidate does not because its required regression half is unavailable. Neither candidate has a controlled intervention or reproducible per-item behavioral source. The frozen breadth scope therefore has one of two required complete audits and zero of one required behavioral benchmarks, so the breadth criterion is not met. This historical decision does not discard either case and is not changed by the later, independently preregistered CLUTRR expansion.

## Appendix H. CLUTRR External Behavioral Validation

### H.1 Frozen Scope and Semantic Identity

Both studies use CLUTRR Task 1 at relation length three, without distractor facts. A row contains three kinship-relation sentences and one queried relation from 23 output classes. Entity names are canonicalized by order of appearance. The semantic ID is

$$(\text{target relation}, \text{edge relation}_1, \text{edge relation}_2, \text{edge relation}_3),$$

where the edges follow the ordered proof chain. Sentence order and entity surface names are excluded from this ID.

For each source story, all six sentence permutations receive one shared `source_group_id`. Eligible targets have at least 45 complete groups. The evaluation rows use the first 34 groups after deterministic sorting: six permutations from groups 0–32 and two from group 33, for 200 rows per target. Exposure uses the next 11 groups: six permutations from groups 34–43 and four from group 44, for 64 rows per target. Thus no source group contributes to both exposure and evaluation.

Targets are paired only when they have the same output label. For each seed, a frozen SHA-256 bit assigns one member of every pair to each arm. The two arms share 22,720 common-base rows with no target semantic ID and append 1,280 exposure rows, giving 24,000 rows per run. Label histograms, total budget, tensor shape, initialization, epoch orders, and evaluation rows are paired.

### H.2 Model, Training, and Primary Analysis

The model is a byte-level Transformer with four layers, model dimension 128, four attention heads, feed-forward dimension 512, dropout 0.1, pooling over a classification token, and 23 output classes. Training uses deterministic CPU float32 execution, AdamW, batch size 128, learning rate $3 \times 10^{-4}$, and 30 fixed epochs. Each epoch contains 188 updates, so the final checkpoint follows 5,640 updates. There is no validation selection or early stopping.

For seed $s$, let $\mathbf{Z}_s^A$ be the complete target-exposure assignment in Arm A and let $\mathbf{Z}_s^B = \mathbf{1} - \mathbf{Z}_s^A$. For target $f$, $\mathbf{Z}_{s,f}^{(1)}$ is whichever arm assignment exposes $f$, and $\mathbf{Z}_{s,f}^{(0)}$ is the complementary assignment. Let $A_{s,f}(\mathbf{z})$ be row-weighted accuracy under complete curriculum assignment $\mathbf{z}$. The primary estimand is

$$\tau_{\mathrm{pp}} = \frac{100}{SF} \sum_{s=1}^{S} \sum_{f=1}^{F} \left[ A_{s,f}(\mathbf{Z}_{s,f}^{(1)}) - A_{s,f}(\mathbf{Z}_{s,f}^{(0)}) \right]. \tag{19}$$

Targets and seeds have equal weight; each target's 200 evaluation rows have equal weight. This target-indexed estimand compares two complete, complementary curricula; it does not identify a one-target direct effect while holding the other target assignments fixed.

The 20,000-replicate percentile bootstrap resamples model seeds, complete same-label semantic-pattern pairs, and source groups within each selected pair member. Exposed and unexposed predictions use the same group draws, and all rows from a sampled group occurrence remain together. Percentiles use NumPy's linear quantile method. A separate analysis code path reconstructs arm alignment from materialized assignments rather than trusting stored arm labels.

The estimand itself is finite-population: it averages the eight frozen seeds and 80 frozen targets. For interval construction, the bootstrap treats the observed seed labels, 40 complete pair labels, and source-group labels within each target as exchangeable under their empirical distributions. The justification is procedural symmetry after freezing: each seed uses the same architecture and training protocol, each selected pair receives the same complementary intervention, and source-group IDs only identify permutations of one source story. Seed and pair resampling preserve their complete clusters; group draws and duplicate seed occurrences are shared across arms. The deterministic target-prefix and source-group allocation prevent a design-based interpretation for all eligible CLUTRR patterns or stories.

The post-hoc sensitivity analysis uses the 640 seed-pattern effects $d_i$. With residual $u_i = d_i - \bar{d}$, each cluster dimension $c$ has CR1 variance

$$\widehat{V}_c = \frac{G_c}{G_c - 1} \frac{1}{N^2} \sum_{g=1}^{G_c} \left( \sum_{i \in g} u_i \right)^2 .$$

Clustering by model seed and complete pattern pair gives $\widehat{V}_{\text{two-way}} = \widehat{V}_{\text{seed}} + \widehat{V}_{\text{pair}} - \widehat{V}_{\text{seed} \times \text{pair}}$. The resulting standard error is 2.6187 points. A conservative two-sided $t_7$ interval is $[17.1469, 29.5313]$ points, still above zero. This post-hoc sensitivity is not a new decision rule or a replacement for the registered hierarchical bootstrap.

### H.3 V3 Pilot and Frozen Precision Decision

V3 contains 20 pattern pairs, 40 targets, eight seeds, and two arms, for 16 runs and 128,000 saved evaluation predictions. Before outcomes, the protocol required any positive, negative, or equivalence conclusion to have a 95% interval no wider than two percentage points.

Table 13: V3 primary result and registered decision.

| Quantity | Value |
|---|---|
| Point estimate | +15.3375 pp |
| 95% percentile interval | [+9.8769, +21.3693] pp |
| Interval width | 11.4924 pp |
| Frozen maximum width | 2.0000 pp |
| Registered outcome | positive but imprecise (precision criterion not met) |

The V3 estimand averages eight frozen seeds and 40 frozen targets. Conditional on that setting, its bootstrap treats those seeds, 20 complete pairs, and within-target source groups as exchangeable empirical draws while preserving clusters and paired arms. Deterministic selection precludes design-based generalization to all eligible CLUTRR patterns or stories.

The interval is directionally positive but fails the registered precision criterion. That decision is retained without reinterpretation.

After V3 was closed, a labeled exploratory analysis used bootstrap-Shapley variance attribution and a seed-by-pair random-effects decomposition. Pattern-pair resampling accounts for 94.40% of bootstrap variance, compared with 4.90% for model seeds and 0.70% for source groups. The random-effects estimates are 162.6402 $\text{pp}^2$ for pair heterogeneity, 0.3490 $\text{pp}^2$ for seeds, and 48.6744 $\text{pp}^2$ for the seed-by-pair interaction.

### H.4 Exploratory Effect Heterogeneity

This exploratory (not preregistered) analysis resolves the V4 contrast into its 80 semantic patterns. The pattern effects have mean 23.34 pp, median 19.4 pp, sample standard deviation 19.8 pp, and range -3.9–75.7 pp; 76/80 are positive. Effect size is strongly associated with unexposed accuracy (Spearman $\rho = -0.847$). A linear regression on that baseline alone has $R^2 = 0.596$; adding target-relation indicators raises it to $R^2 = 0.691$. The median

normalized gain is 0.714, so the raw spread is not only a scale artifact, although available headroom explains much of it.

Mean effects are 25.75 pp when the target relation is absent from the three edges (50 patterns) and 19.32 pp when it is present (30), and 13.31, 22.90, and 23.69 pp for one, two, and three distinct edge types (1, 22, and 57 patterns). Neither covariate predicts the effect at conventional levels (two-sided Spearman $p = 0.188$ and $p = 0.931$, respectively). Target relation as a categorical grouping explains 0.494 of the effect variance, but leave-one-relation mean effects remain within 20.91–24.64 pp. Figure 4 shows the full distribution. Exposure gains are largest where the unexposed model is far from ceiling, and normalized gain is high for most patterns; the heterogeneity is thus mostly a headroom effect rather than evidence for a particular compositional mechanism.

A calibrated prospective projection finds that the original two-point width would require about 634 independent pairs even with infinitely many seeds. Only 139 eligible V3-disjoint pairs remain, and using all of them with infinitely many seeds projects to a 4.2690-point width. The strict-precision replication route was therefore closed rather than consuming more seeds under an infeasible criterion.

### H.5 Prospective V4 Design

V4 treats V3 as a declared pilot and changes the scientific question from a two-point precision target to directional confirmation. The prospective design effect is

$$\min(0.5 \times 15.3375,\ 9.8769) = 7.66875 \text{ pp.}$$

A 100,000-draw 64-bit permuted congruential generator (PCG64) Monte Carlo simulation evaluates one to four 20-pair blocks. The frozen selection rule chooses the smallest block count for which estimated power and the 95% Wilson lower confidence bound are both at least 0.90.

For candidate pair count $Q \in \{20, 40, 60, 80\}$, set the frozen design effect to $\delta = 7.66875$ points and draw

$$\widehat{\tau}_Q = \delta + \mathrm{SE}_Q Z, \qquad Z \sim N(0,1),$$
$$\mathrm{SE}_Q = c\sqrt{\frac{\sigma_s^2}{8} + \frac{\sigma_p^2}{Q} + \frac{\sigma_{sp}^2}{8Q}}.$$

The V3 random-effects inputs are $\sigma_s^2 = 0.3490014$, $\sigma_p^2 = 162.6401609$, and $\sigma_{sp}^2 = 48.6744361$ pp$^2$, with frozen calibration $c = 1.0067880$. The seed effect is shared across pairs, the pair effect across seeds, and the interaction is seed-by-pair. Source groups are nested within targets and shared across arms; their contribution is not a fourth explicit random effect, but enters $c$ through calibration to the full V3 hierarchical percentile width, which resampled source groups. A simulated experiment succeeds exactly when $\widehat{\tau}_Q \geq 1$ and $\widehat{\tau}_Q - 1.959964\,\mathrm{SE}_Q > 0$. Power is the fraction of 100,000 successful draws; the block-size rule additionally requires its binomial 95% Wilson lower endpoint to be at least 0.90.

Table 14: V4 power analysis for the first two candidate sizes.

| Blocks | Pairs | Runs | Power | Wilson lower |
|---|---|---|---|---|
| 1 | 20 | 16 | 0.74397 | 0.74126 |
| 2 | 40 | 32 | 0.95828 | 0.95702 |

The minimum-passing design therefore has two blocks, 40 pairs, 80 patterns, and 32 runs. All target semantic IDs are disjoint from V3. The primary success criterion, frozen before materialization and training, requires $\tau_{\text{pp}} \geq 1$ and a two-sided 95% interval lower endpoint above zero.

### H.6 V4 Execution Integrity and Result

Materialization validates 80 targets, 16,000 evaluation rows across two blocks, 5,120 exposure rows, the shared 22,720-row common base, and all 32 training manifests. The frozen execution assets include a $43{,}840 \times 185$ tensor cache, 32 arrays of 24,000 training indices, two 8,000-row evaluation caches, eight initialization records, and 240 epoch-order records.

All 32 runs reach epoch 30. The post-execution validator checks finite model and optimizer state, 960 run-by-epoch order records, 256,000 predictions, four-way RNG pairing across both blocks and arms within each seed, prediction ordering, separate-code-path metric reconstruction, and artifact hashes.

Table 15: V4 primary and descriptive results. Sign counts do not define additional hypothesis tests.

| Quantity | Value |
|---|---|
| Point estimate | +23.3391 pp |
| 95% percentile interval | [+17.7498, +29.2257] pp |
| Registered outcome | directional criterion met |
| Block effects | +25.0109, +21.6672 pp |
| Positive seeds | 8/8 |
| Positive pairs | 40/40 |
| Positive target patterns | 76/80 |
| Cross entropy, exposed minus unexposed | −1.4938 |
| Brier score, exposed minus unexposed | −0.4224 |
| ECE-15, exposed minus unexposed | −0.2018 |

The V4 point estimate exceeds one point and its interval lies above zero, so the registered directional positive rule passes. V3 and V4 are not pooled: they use disjoint targets, answer different registered decision questions, and retain separate bootstrap draws and classifications.

### H.7 Paper Artifact

The compact CLUTRR bundle contains 51 checksum-managed files: design and effect preregistrations, freeze validations, execution-integrity reports, run-artifact hashes,

exact bootstrap draws, pair/pattern/seed tables, code snapshots, claim boundaries, source-file provenance, and a standalone verifier. It excludes local paths and large checkpoints. Its `SHA256SUMS` file has SHA-256 `210dddc02840a7bab7fe1215378d36ac86cc574a06aeca238482eab86ba3ef36`.

## Appendix I. Exploratory Semantic Versus Surface Exposure

This exploratory (not preregistered) intervention uses the 40 block-0 CLUTRR patterns and their shared 8,000-row evaluation set. Every arm appends 48 relation-matched rows per pattern to the same 22,720-row common base, giving 24,640 training rows with identical label counts. Arm C0 uses semantically unrelated filler rows. Arm C1 uses eight complete source-story groups with the target semantic identity. Arm C2 uses distinct semantic identities with the same target relation and at least two matching edge types as a multiset. All patterns met this strict two-edge criterion; no one-edge fallback or main-analysis exclusion was needed.

The four seeds share initialization and epoch order across arms. The model, optimizer, and 30-epoch budget match the V4 configuration; execution uses deterministic float32 Metal Performance Shaders. The protocol was frozen before training at hash prefix `0fb766ba884f`; its one-epoch timing run took 32.74 seconds. Pattern-cluster intervals use 2,000 resamples with seed zero, and seed intervals are paired $t_3$ intervals.

The mean arm accuracies are 60.3% for C0, 82.8% for C1, and 64.0% for C2.

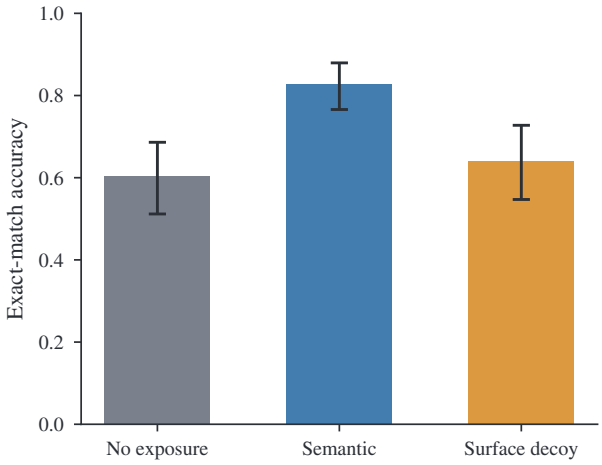



Figure 6: Exploratory accuracy under no target exposure, same-semantic exposure, and surface-similar decoys. Bars show mean accuracy; error bars are 95% pattern-cluster intervals.

The pattern interval for C2–C0 is positive, although its four-seed $t$ interval includes zero, so the decoys are not treated as having exactly no effect. More importantly, C1 exceeds C2 by 18.8 pp with pattern interval [13.2, 25.3]. Under the decision rule frozen in the protocol before training (not publicly preregistered), this supports the interpretation that semantic identity, not surface similarity alone, is the primary source of the gain in this intervention.

The result remains bounded to these patterns, decoy criteria, architecture, and training budget.

## Appendix J. Certified Exposure Intervals

### J.1 Assumptions and Proposition

Fix one semantic namespace and evaluation identities $E = \{e_1, \ldots, e_N\}$. Let $z_i = 1$ if evaluation identity $e_i$ occurs semantically in training, and $z_i = 0$ otherwise. Assume: (A1) the finite denominator is frozen before inspection; (A2) every identity has a deterministic canonical semantic ID; (A3) each executable certificate supplies a sound constraint $\ell_j \leq \sum_{i \in G_j} z_i \leq u_j$; (A4) certificate groups are pairwise disjoint; and (A5) uncovered identities receive no constraint beyond binary membership. Appendix J.4 relaxes A4 computationally: mixed-integer linear programming gives exact bounds for tractable overlapping systems, while a linear-programming relaxation gives a dual-certified outer interval.

**Proposition 1 (restated).** Let $R = E \setminus \cup_j G_j$. Then

$$\sum_j \ell_j \leq C(E) := \sum_{i=1}^{N} z_i \leq |R| + \sum_j u_j. \tag{20}$$

Both endpoints are attained by some exposure assignment consistent with the certificate system, so no tighter interval follows from the certificates alone.

**Proof.** Disjointness partitions the total:

$$C(E) = \sum_j \sum_{i \in G_j} z_i + \sum_{i \in R} z_i.$$

Applying each lower certificate and $\sum_{i \in R} z_i \geq 0$ gives the lower bound. Applying each upper certificate and binary membership $\sum_{i \in R} z_i \leq |R|$ gives the upper bound. To attain the endpoints, choose exactly $\ell_j$ exposed identities in each $G_j$ and none in $R$; this feasible assignment attains the lower endpoint. Choosing exactly $u_j$ in each group and every identity in $R$ attains the upper endpoint. No tighter universal interval follows from these certificates alone. □

### J.2 Grouped-Seed Corollary

For a fixed seed $s$, suppose $G_s = \{\phi(1, s), \ldots, \phi(K, s)\}$ contains all evaluated categories, every training episode using $s$ selects exactly one member of $G_s$, and $s$ appears $m_s$ times. If a complete signature census excludes cross-seed aliases, then $m_s = 0$ gives group constraint $[0, 0]$, while $m_s > 0$ gives

$$1 \leq \sum_{i \in G_s} z_i \leq \min(K, m_s). \tag{21}$$

The lower bound follows because at least one category was selected; the upper bound follows because at most $K$ categories and at most $m_s$ distinct episode assignments can occur. Persisting category assignments would collapse this interval to the exact number of observed categories.

### J.3 Novelty-Conditioned Score Interval

Let $\mathcal{R}$ contain $M$ unresolved items with observed scores $y_i \in [0,1]$, and let the unknown truly novel subset be $\mathcal{N} \subseteq \mathcal{R}$. Suppose external information only implies $|\mathcal{R} \setminus \mathcal{N}| \leq U < M$, and write the ordered scores as $y_{(1)} \leq \cdots \leq y_{(M)}$.

**Proof.** Fix $n = |\mathcal{N}|$. Among all subsets of size $n$, replacing any selected score by a smaller unselected score cannot increase the mean. Repeating this exchange shows that the minimum is the mean of the first $n$ order statistics. The symmetric exchange gives the maximum as the mean of the last $n$ order statistics.

The mean of the first $n$ scores is nondecreasing in $n$: the next score $y_{(n+1)}$ is no smaller than any of the first $n$ and hence no smaller than their mean. Because the exposure constraint permits $n \in \{M-U, \ldots, M\}$, the global minimum is therefore attained at $n = M-U$. The corresponding upper-tail mean is nonincreasing in $n$, so the global maximum is also attained at $n = M-U$. Thus

$$L(U) = \frac{1}{M-U} \sum_{k=1}^{M-U} y_{(k)}, \qquad H(U) = \frac{1}{M-U} \sum_{k=U+1}^{M} y_{(k)}.$$

Taking $\mathcal{N}$ to be the first or last $M-U$ ordered items attains the respective endpoint while excluding exactly $U$ items. Increasing $U$ removes an upper-tail item from the lower construction or a lower-tail item from the upper construction, so $L(U)$ is nonincreasing and $H(U)$ is nondecreasing. At $U = 0$, both equal the observed mean. □

For binary scores with $k$ correct items and $n = M-U$, this reduces to

$$L(U) = \frac{\max(0, k-U)}{n}, \qquad H(U) = \frac{\min(k, n)}{n}.$$

The exposure upper endpoint from Proposition 1 can be used as $U$ when it applies to the unresolved set and is smaller than $M$. If the available upper bound permits all $M$ items to be exposed, a novel mean need not exist; conditional on a nonempty novel set, the interval is $[\min_i y_i, \max_i y_i]$. The construction is a finite trimming argument related to Lee (2009) and the contaminated-data bounds of Horowitz and Manski (1995).

**Parameterized SLR sensitivity.** The structural audit leaves $M = 548$ items unresolved. We use only this denominator, not model predictions or partial scores. Let $p = k/M$ be an unspecified correct fraction. For integer $U < M$,

$$L(p, U) = \frac{\max(0, pM - U)}{M-U}, \qquad H(p, U) = \frac{\min(pM, M-U)}{M-U}.$$

For finite binary scores, $pM$ must be an integer; this condition preserves endpoint attainment. The bounds express a hypothetical sensitivity calculation, not an observed accuracy or an estimated exposure rate.

Writing $\rho = U/M$, the greatest possible width over admissible $p$ is

$$W_{\max}(U) = \max_{k \in \{0,\ldots,M\}} \{H(k/M, U) - L(k/M, U)\} = \min\left(1, \frac{U}{M-U}\right) = \min\left(1, \frac{\rho}{1-\rho}\right).$$

This equality holds for this even $M$: $k = M/2$ attains the maximum. Indeed, the interval width is at most both 1 and $U/(M-U)$. At $k = M/2$, its numerator is $U$ when $U \le M/2$ and $M - U$ when $U \ge M/2$, attaining the respective bound. Figure 7 therefore shows how much identification can deteriorate as the assumed exposure upper bound increases, without choosing a model accuracy.

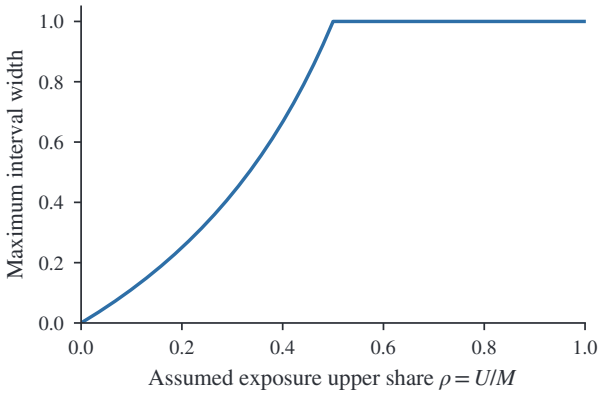



Figure 7: Parameterized sensitivity for the $M = 548$ structurally unresolved SLR-Bench items. The curve shows the maximum interval width over hypothetical binary correct fractions as the assumed exposure upper share varies. No model scores are used; width describes partial identification, not sampling uncertainty.

### J.4 Overlapping Certificates

For overlapping groups, let $A_{ji} = 1$ when item $i$ belongs to certificate group $G_j$. The compatible exposure assignments solve

$$\min / \max \; w^\top z \quad \text{subject to} \quad \ell \le Az \le u, \qquad z \in \{0,1\}^N.$$

The toolkit function `solve_overlapping_certificate_bounds` solves this binary program by mixed-integer linear programming for systems with at most 200 items. Larger systems, or an explicit `lp` request, replace binary membership by $z \in [0,1]^N$. The resulting interval is an outer interval and is labeled as such rather than being reported as attained by a binary assignment.

Each relaxed endpoint includes an independently checkable weak-duality certificate. For the lower endpoint, nonnegative $(\alpha, \beta, \gamma)$ satisfying

$$A^\top(\beta - \alpha) - \gamma \le w$$

certifies

$$\ell^\top \beta - u^\top \alpha - \mathbf{1}^\top \gamma \le \min w^\top z.$$

The upper endpoint applies the same check to $-w$. A separate verifier checks dimensions, signs, feasibility, and the claimed objective within $10^{-9}$.

The relaxation can be strict. With three items and exact-one constraints on $\{1, 2\}$, $\{2, 3\}$, and $\{1, 3\}$, no binary assignment is feasible, while $z = (1/2, 1/2, 1/2)$ satisfies the relaxation and gives interval $[1.5, 1.5]$. Focused deterministic properties compare 100 random small systems with exhaustive enumeration and independently verify 100 generated dual certificates, in addition to this odd-cycle case and the existing disjoint-certificate regression tests.

### J.5 Falsification and Design Consequences

The disjoint closed-form function still rejects overlapping certificate groups, preserving its existing contract; the new solver handles those systems explicitly. Both interfaces reject incoherent bounds and out-of-range indices. An assignment checker directly falsifies a certificate if observed identities violate its interval.

The proposition changes reporting and design in four ways: report $[L, U]$, never code no-witness rows as novel; rerun witnesses after split mutation; persist semantic IDs and category assignments for online generators; and keep exposure bounds separate from behavioral estimands.

## Appendix K. Portable Toolkit Validation

The versioned `semantic-audit-v1` command exposes Tensor Logic and Occam adapters through one schema. Required fields include source revisions and hashes, observation contracts and semantic-unit adequacy rules, train/evaluation identities, semantic IDs, witness types, completeness, overlap denominators, behavioral metadata, and claim wording. The validator enforces that exact evidence has a Boolean decision, sound lower bounds establish only positive overlap, unresolved evidence asserts neither overlap nor novelty, and every current adapter declares its observable semantics.

Table 16: Benchmark-level implementation assurance. Residual risk is kept separate from identification bounds; no row implies a quantified implementation-error rate.

| Case | Semantic-ID implementation | Source traceability | Soundness and validation coverage | Known failure mode | Residual risk |
|---|---|---|---|---|---|
| Internal finite rules | Complete 20-output vector | Frozen generator, split, stream, and checkpoint hashes | Exhaustive finite-map census; paired-manifest and metric recomputation | Applies only to the finite deterministic domain | Low conditional risk; implementation error is not rate-bounded |
| Boolean tasks | Signed-literal vector, active mask, or literal subset | Pinned released generators and full stream manifests | Combinatorial injectivity plus separate-code-path collision and replacement checks | Exactness is limited to the three selected released task definitions | Low within those definitions; transfer to conflicting definitions is unsupported |
| CLUTRR | Ordered target and three-edge relation tuple | Frozen CLUTRR extraction, source groups, manifests, and artifact hashes | Deterministic extraction; source-group disjointness; 11 post-training integrity classes and separate-code-path metric recomputation | Task 1, length three, and the frozen relation parser only | Unquantified parser risk; low within the validated materialization |
| SLR-Bench | Token identity plus restricted-fragment canonicalizer | Pinned `v1-All` release and row-level train–test witnesses | Each reported semantic pair has a proof-producing witness; positive and negative normalizer tests | Aggregates, negation, arithmetic, and unresolved disjunction are not decided | Low false-positive risk for reported witnesses; unknown missed overlap |
| Tensor Logic | Endpoint-path and ordered-relation-schema IDs | Pinned repository and graph artifacts | Exhaustive post-scrub path re-enumeration; adapter canaries and schema tests | No checkpoint, per-query predictions, or training logs | Exposure remains an interval; behavioral impact is unresolved |
| Occam Markov | Exact transition-tensor ID with grouped-seed certificates | Pinned repository and enumerated seed range | 75,093-signature census; bound checker and grouped-seed tests | Order assignments are unpersisted; regression code is absent | Interval retained for Markov tasks; regression scope remains unresolved |

**Validator independence dimensions.** We use no unqualified independence claim: a validator can differ from a producer by implementation, code path, execution environment, operator, or source specification, and "separate-code-path recomputation" here means a different analysis or checking routine consuming frozen producer artifacts, while "isolated reproduction" means a fresh process and dependency environment. Every major claim is recomputed on such a separate path: the internal exact-alias effect from paired manifests and saved predictions; the Boolean exposure and effect from collision-replacement checks and cluster-effect recomputation; the CLUTRR V4 effect by `train_integrity.py` plus assignment and metric reconstruction; the SLR lower bound by row-level witness checks with positive canaries and negative normalizer tests; the Tensor Logic and Occam outputs by a common schema validator with two isolated reproductions; and the certified interval by an assignment checker over exhaustive finite cases. All of these remain in the same project and share repository, protocol, pinned sources, and operator, so they limit but never eliminate correlated implementation error; we claim neither an independent human operator nor an independently authored specification.

Eleven tests cover the three required canaries, schema rejection, and both public adapters. Two isolated `python3 -I` reproductions use no third-party runtime package and produce byte-identical adapter outputs, manifest, and checksums. All portability checks pass; they validate the toolkit and do not change the original breadth decision.

## Appendix L. Reproducibility Map

Table 17: Claim-to-artifact map. Paths are relative to the audit supplement root supplied with the manuscript. The public artifact repository is `https://github.com/zihanzhou0725/semantic-novelty-audit-artifact`.

| Claim or artifact | Reproduction source |
|---|---|
| Internal exact audit (historical p1) | `analysis/quality-enhancement-v1/p1/results.json` and frozen paired manifests |
| SLR lower bound | `artifacts/slr_overlap_results.json`, token witnesses, and semantic witnesses |
| SLR identification decision (historical p2) | `analysis/quality-enhancement-v1/p2/identification_report.json` |
| Boolean coverage | `protocols/boolean-behavioral-v1/split_manifest.json` |
| Boolean behavior | `analysis/boolean-behavioral-v1/results.json` and paired episode CSV |
| Prospective external audits (historical p3) | `analysis/quality-enhancement-v1/p3/` and candidate manifest |
| CLUTRR V3 pilot and diagnosis | `results/v3/`, `design/v3_variance_results.json`, and frozen protocols |
| CLUTRR V4 confirmation | `results/v4/`, `design/v4_design_manifest.json`, frozen protocols, and `analysis/external-behavior-v4/cluster-sensitivity-v1/` |
| Portable toolkit (historical p4) | `analysis/quality-enhancement-v1/p4/` and `toolkit/semantic_audit/` |
| Certified interval (historical p5) | `analysis/quality-enhancement-v1/p5/` and executable bounds tests |
| Integrity | protocol/full-matrix validators and frozen checksums |

From the manuscript directory, `make reproduce-paper` regenerates the five strengthening analyses from retained evidence, copies their macros and figures, regenerates the overview, and runs `latexmk`. Each analysis writes a `summary.json` and a `run_info.json` recording its inputs under `analysis/tmlr-strengthening-v1/`: T2 (CLUTRR heterogeneity), T3a (internal heterogeneity), T6 (semantic versus decoy, with compact predictions in `artifacts/tmlr_t6_run_outputs/`), T9 (overlapping bounds, alongside `toolkit/semantic_audit/bounds.py`), and T10 (parameterized sensitivity). Set `PYTHON` to an environment with the supplement dependencies and `AUDIT_ROOT` to the audit supplement directory if it is not in the default sibling layout. No model training is required.